

Appel à candidature – Doctorat CIFRE

Title: *Impact of Knowledge Graph Quality on Trustworthy AI Systems*

Mots-clés : Trustworthy AI, Knowledge Graph Quality, Knowledge Graph Completion, Large Language Models, Retrieval-Augmented Generation (RAG).

Background and Motivation

Large Language Models (LLMs) and AI Agents are rapidly transforming enterprise decision-support systems. Recent advances such as Retrieval-Augmented Generation (RAG) [1] and GraphRAG [2] combine LLMs with enterprise Knowledge Graphs (KGs) to improve factual grounding, contextual reasoning, and explainability. Significant progress has also been made in knowledge graph completion [4,9], while evaluation frameworks such as HELM [3] and LLM-as-a-Judge [5,6] have considerably improved the evaluation of LLM-generated outputs.

Despite these advances, enterprise knowledge remains incomplete and continuously evolving. Missing entities, relationships, or business rules may directly affect AI reasoning and lead to unreliable recommendations. Existing approaches mainly focus on improving knowledge retrieval, completing missing graph links, or evaluating generated outputs. However, they still assume that the knowledge available to the AI system is sufficient before reasoning takes place. As a result, AI systems may produce convincing recommendations even when critical knowledge is missing.

This PhD addresses the following research question: How can an AI system determine whether the available knowledge is sufficiently complete to produce reliable recommendations?

The project is based on the hypothesis that the reliability of AI reasoning depends not only on the reasoning model itself but also on the completeness of the knowledge available before reasoning begins. The objective is to develop methods for assessing knowledge graph completeness, identifying critical knowledge gaps, and investigating how knowledge completeness influences the reliability of AI reasoning and the confidence associated with AI-generated recommendations.

Context and Partnership

The CIFRE PhD will be carried out at LIPADE (Université Paris Cité), under the supervision of Soror Sahri, whose research focuses on data and knowledge quality management. Industrial supervision will be provided by SpotWorkAI, under the supervision of Karim Toumi.

SpotWorkAI develops AI-based decision-support assisting experts in complex decision-making processes. These systems increasingly rely on enterprise knowledge graphs and Large Language Models to exploit heterogeneous organizational knowledge. Although these technologies have considerably improved information retrieval and contextual reasoning, their industrial deployment highlights a fundamental limitation: current AI systems cannot determine whether the knowledge available before reasoning is sufficiently complete to support reliable recommendations.

This limitation represents a major industrial challenge for SpotWorkAI. In real operational environments, enterprise knowledge continuously evolves and is distributed across multiple heterogeneous repositories. Missing or incomplete knowledge may directly impact AI reasoning while remaining undetected by existing GraphRAG or AI Agent architectures. Improving reasoning reliability therefore requires mechanisms capable of assessing the completeness of enterprise knowledge before recommendations are generated.

This collaboration provides a unique research environment combining academic expertise with real industrial datasets, enterprise knowledge bases, and operational feedback, enabling the proposed methods to be validated under realistic conditions while facilitating their transfer to enterprise AI systems.

Scientific Objectives

This PhD will investigate methods for assessing the completeness of enterprise Knowledge Graphs for AI reasoning. It will study how missing knowledge influences reasoning quality and recommendation reliability, and develop methods for automatically identifying critical knowledge gaps. The project will also investigate different strategies for enriching the knowledge available to AI systems, including GraphRAG, Knowledge Graph Completion, and external knowledge sources. Finally, the proposed methods will be evaluated using task-specific benchmarks together with recent LLM-as-a-Judge approaches [7,8] and validated on real-world decision-support applications provided by SpotWorkAI.

Expected Candidate Profile

Applicants should hold an MSc (or equivalent) in Computer Science, Artificial Intelligence, Data Science, or a closely related discipline. A strong background in machine learning and large language models is expected, together with an interest in knowledge graphs, and Trustworthy AI. Experience with Python and modern AI frameworks is desirable, as well as good scientific communication skills in English.

Conditions and modalities

Contract type: CIFRE PhD position (3 years)

Employer: SpotWorkAI

Place of work: LIPADE (Université Paris Cité) & SpotWorkAI

Expected start date: Ideally before the end of 2026

Gross annual salary: €27,600

References

- [1] Lewis et al., Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks, NeurIPS, 2020.
- [2] Edge et al., From Local to Global: A GraphRAG Approach to Query-Focused Summarization, Microsoft Research, 2024.
- [3] Liang et al., Holistic Evaluation of Language Models (HELM), arXiv:2211.09110.
- [4] Zhong et al., A Comprehensive Survey on Automatic Knowledge Graph Completion, ACM Computing Surveys, 2023.
- [5] Liu et al., G-Eval: NLG Evaluation using GPT-4 with Better Human Alignment, ACL, 2023.
- [6] Kim et al., Prometheus 2: An Open Source Language Model Specialized in Evaluating Other Language Models, ACL, 2024.
- [7] Xu et al., Towards Trustable Language Models: Investigating Information Quality of Large Language Models, arXiv:2401.13086. 2024.
- [8] Towards Trustworthy Knowledge Graph Reasoning, AAAI, 2025.
- [9] Sahri et al. OMNIA: Closing the Loop by Leveraging LLMs for Knowledge Graph Completion. arXiv:2603.11820.